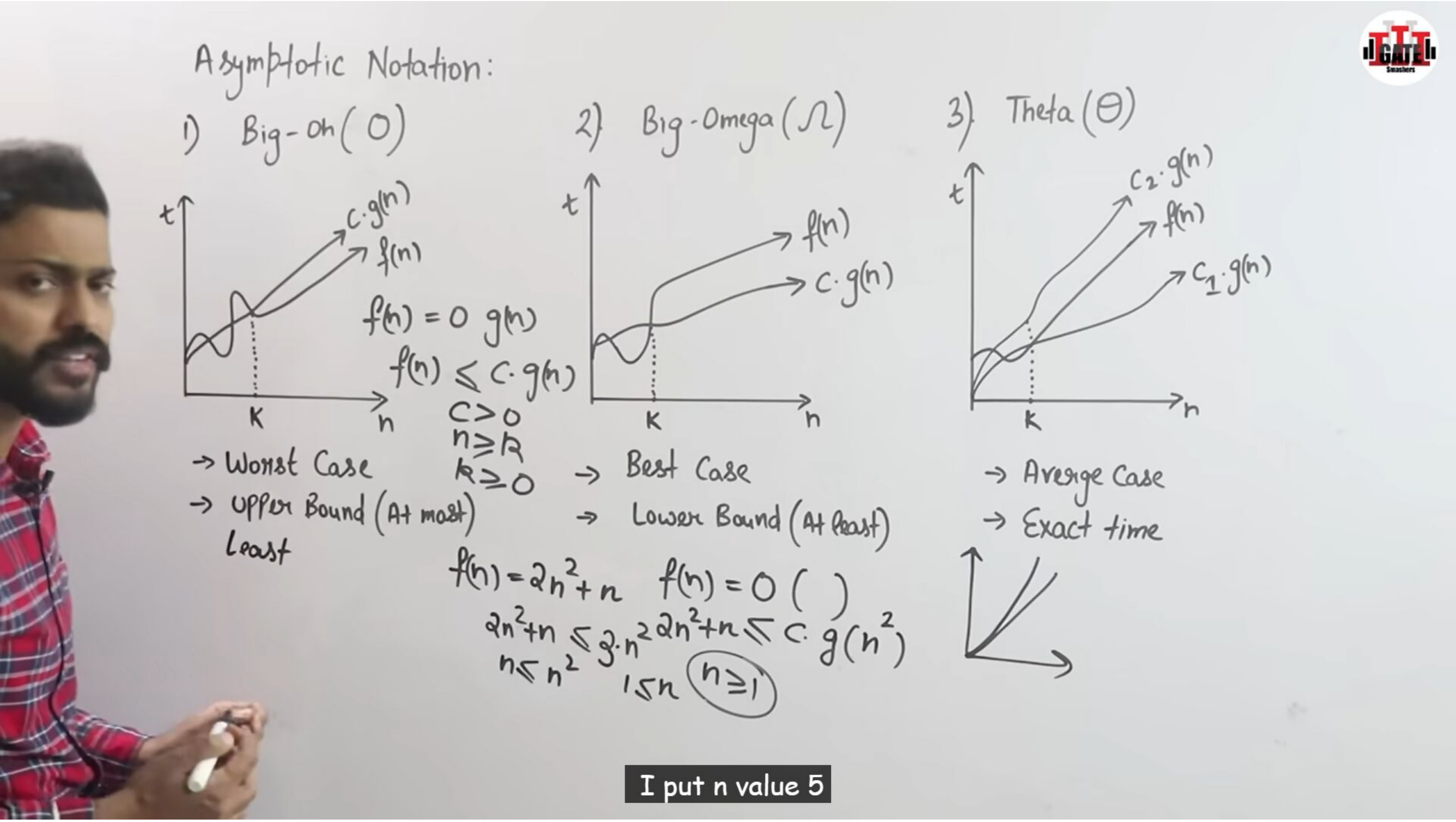
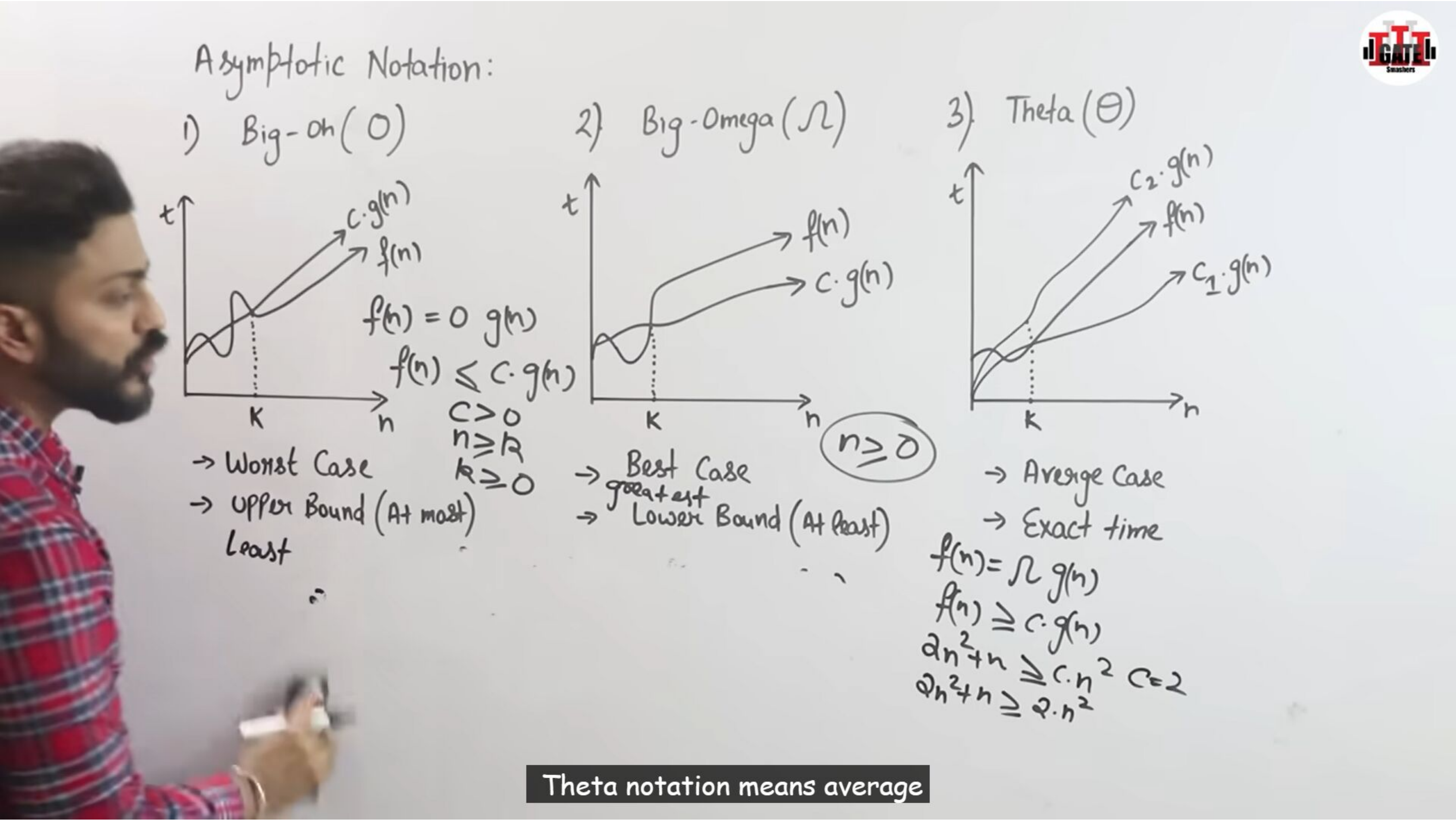

Asymptotic Notation:
1) Big-Oh (O)
2) Big-Omega (Ω)
3) Theta (Θ)
t
c·g(n)
f(n)
K
n
f(n) = O g(n)
f(n) ≤ C·g(n)
C > 0
n ≥ k
k ≥ 0
→ Worst Case
→ Upper Bound (At most)
Least
f(n)
c·g(n)
→ Best Case
→ Lower Bound (At least)
c2·g(n)
f(n)
c1·g(n)
→ Avergе Case
→ Exact time
f(n) = 2n² + n
f(n) = O ( )
2n² + n ≤ 3·n²
2n² + n ≤ C·g(n²)
n ≤ n²
1 ≤ n
n ≥ 1
I put n value 5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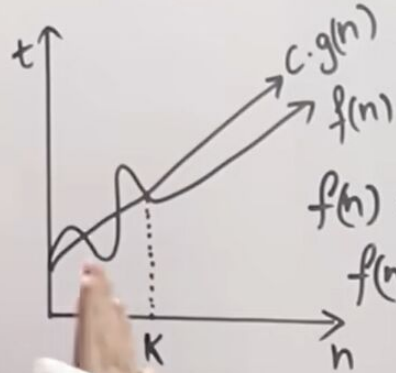
f(n)
c1·g(n)
K
n
→ Averge Case
→ Exact time
f(n) = Ω g(n)
f(n) ≥ c·g(n)
2n²+n ≥ c.n² c=2
2n²+n ≥ 2.n²
Theta notation means average

Asymptotic Notation:

1) Big-Oh (O)



$f(n) = O\ g(n)$

$f(n) \leq c \cdot g(n)$

$c > 0$

$n \geq k$

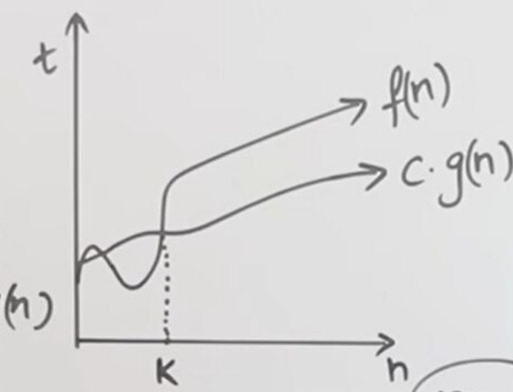
$k \geq 0$

– …st Case

– …per Bound (At most)

…east

2) Big-Omega (Ω)



$n \geq 0$

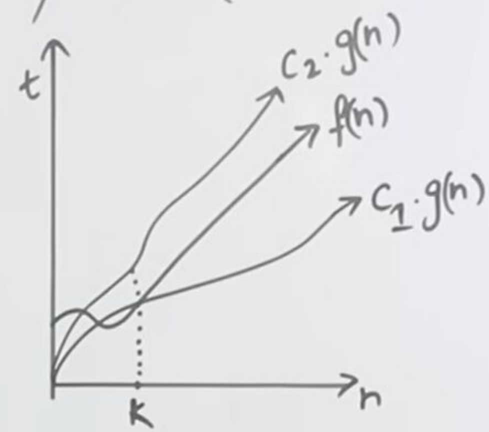
→ Best Case

→ greatest Lower Bound (At least)

$c_1 \cdot g(n) \leq f(n) \leq c_2 \cdot g(n)$

$2n^2 \leq 2n^2 + n \leq 3n^2$

3) Theta (Θ)



→ Averge Case

→ Exact time

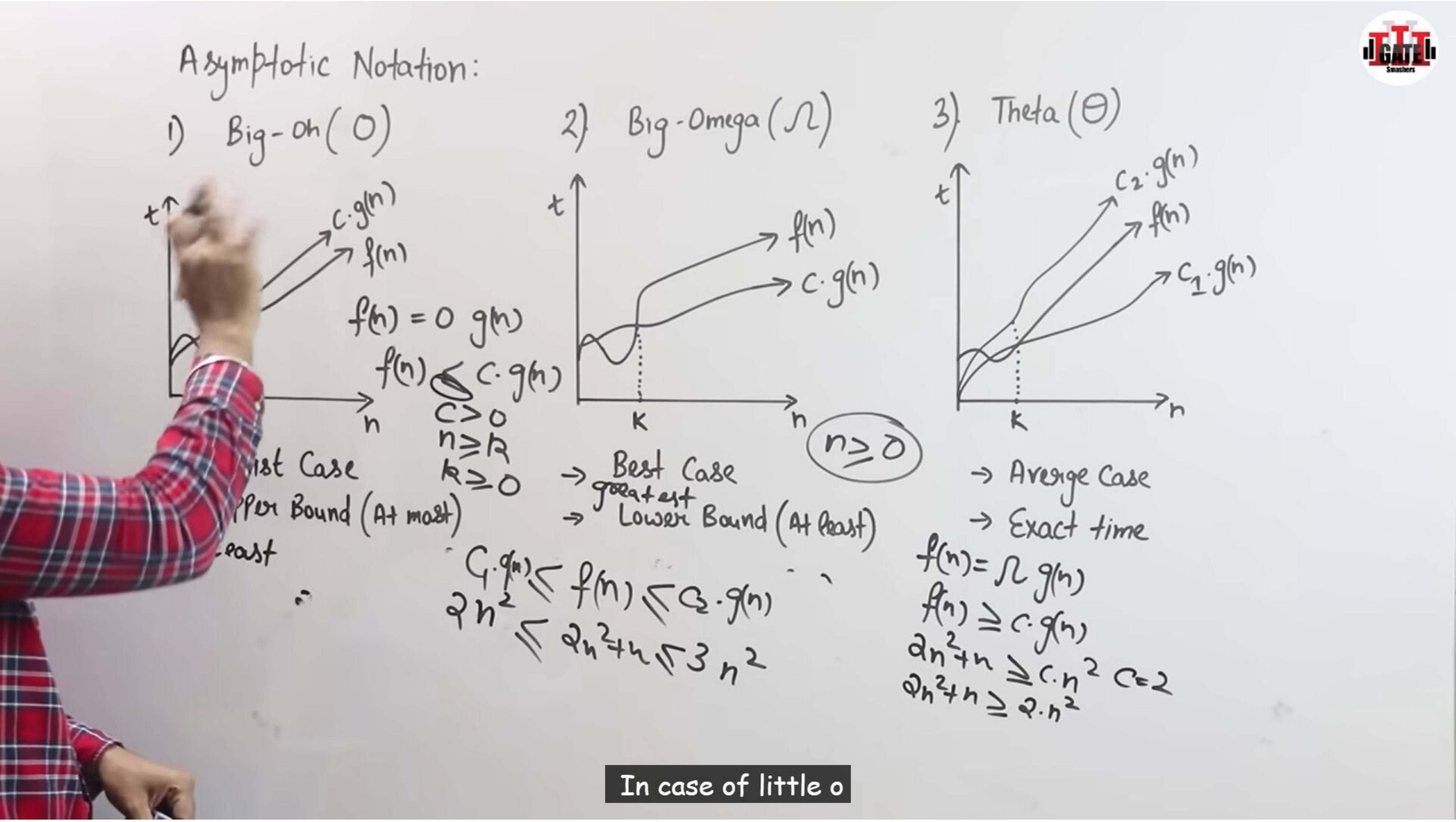
$f(n) = \Omega\ g(n)$

$f(n) \geq c \cdot g(n)$

$2n^2 + n \geq c \cdot n^2 \quad c = 2$

$2n^2 + n \geq 2 \cdot n^2$

So this is how we represent the big O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K
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greatest
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C1.g(n) ≤ f(n) ≤ C2.g(n)
2n² ≤ 2n²+n ≤ 3n²
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f(n) = Ω g(n)
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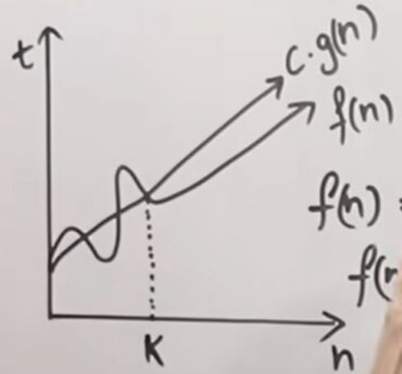
2n²+n ≥ c.n² c=2
2n²+n ≥ 2.n²
GATE Smashers
In case of little o

Asymptotic Notation:

1) Big-Oh (O)



$f(n) = O\ g(n)$

$f(n) \leq c\ g(n)$

> 0

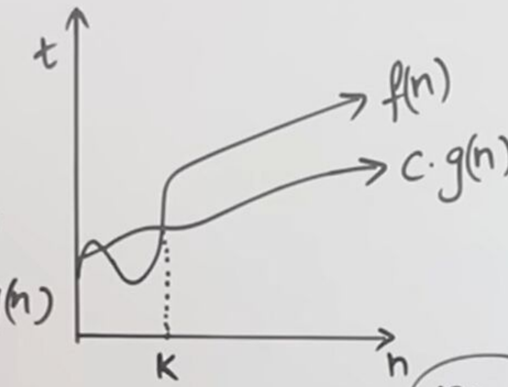
$\geq k$

$k \geq 0$

→ Worst Case

→ ... most)

2) Big-Omega (Ω)



$n \geq 0$

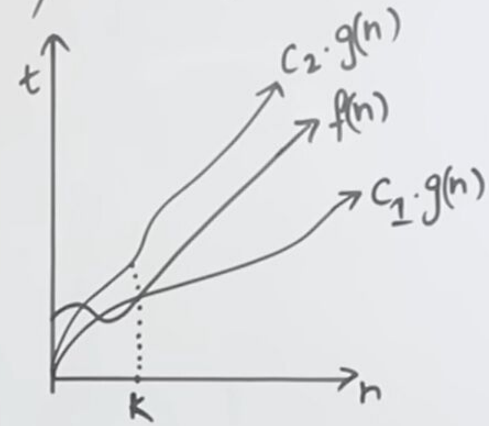
→ Best Case

→ greatest Lower Bound (At least)

$c_1 \cdot g(n) \leq f(n) \leq c_2 \cdot g(n)$

$2n^2 \leq 2n^2 + n \leq 3n^2$

3) Theta (Θ)



→ Averge Case

→ Exact time

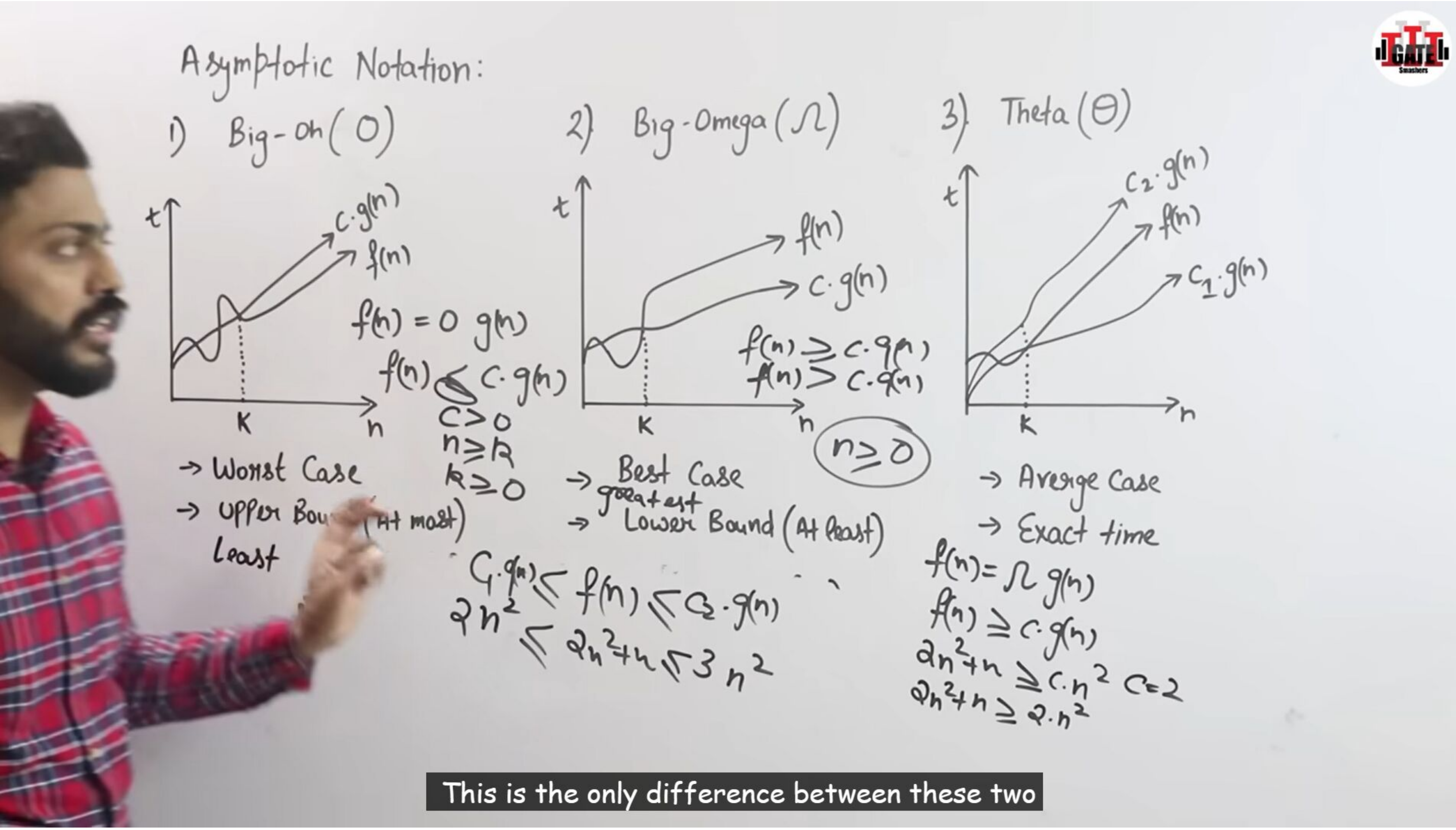
$f(n) = \Omega\ g(n)$

$f(n) \geq c \cdot g(n)$

$2n^2 + n \geq c \cdot n^2 \quad c=2$

$2n^2 + n \geq 2 \cdot n^2$

It does not come equal to, rest of the funda is same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2n² ≤ 2n²+n ≤ 3n²
c₂·g(n)
c₁·g(n)
→ Averge Case
→ Exact time
f(n) = Ω g(n)
f(n) ≥ c·g(n)
2n²+n ≥ c·n² c=2
2n²+n ≥ 2·n²
This is the only difference between these two